Transformers — Solution

X26 (2026) — English translation

A1 **0.20 pts**

What are the magnetic field fluxes through the primary and secondary windings Φ_1 and Φ_2? Express your answers using Φ, N_1 and N_2.

Answer: Answer:

$$\Phi_1 = \Phi \cdot N_1, \qquad \Phi_2 = -\Phi \cdot N_2$$

A2 **0.20 pts**

Using the circulation theorem for the magnetic field strength vector $\vec{H}$, find the magnetic field flux Φ in the core. Express your answer in terms of S, ℓ, μ, I_1, I_2, N_1 and N_2.

Let us write the circulation theorem for the entire transformer:

$$H \cdot l = N_1 I_1 - N_2 I_2,$$

Let us express the magnetic field flux:

Answer: Answer:

$$\Phi = BS = \mu\mu_0 HS = \frac{\mu\mu_0 S}{l}(N_1 I_1 - N_2 I_2)$$

A3 **0.30 pts**

Obtain expressions for the inductances of windings L_1, L_2 and their mutual inductance coefficient M. Express your answer using S, ℓ, μ, I_1, I_2, N_1 and N_2.

Answer: Answer:

$$L_1 = \frac{\partial \Phi_1}{\partial I_1} = \frac{\mu\mu_0 S N_1^2}{l}$$
$$L_2 = \frac{\partial \Phi_2}{\partial I_2} = \frac{\mu\mu_0 S N_2^2}{l}$$
$$M = \frac{\partial \Phi_1}{\partial I_2} = \frac{\partial \Phi_2}{\partial I_1} = -\frac{\mu\mu_0 S N_1 N_2}{l}$$

A4 **0.40 pts**

Find the magnetic induction emf $\mathcal{E}_1$ and $\mathcal{E}_2$ arising in the primary and secondary windings, respectively. Express your answer using $\dot{I}_1, \dot{I}_2, L_1, L_2$ and M. Notes: EMF $\mathcal{E}_1$ is considered positive if the electrostatic potential at the top point a1 is greater than at the bottom a2. EMF $\mathcal{E}_2$ is considered positive if the electrostatic potential at the top point b1 is greater than at the bottom b2.

Find the magnetic induction emf $\mathcal{E}_1$ and $\mathcal{E}_2$ arising in the primary and secondary windings, respectively. Express your answer using $\dot{I}_1, \dot{I}_2, L_1, L_2$ and M.

Notes:

Answer: Answer:

$$\mathcal{E}_1 = \dot{\Phi}_1 = N_1 \dot{\Phi} = L_1 \dot{I}_1 + M \dot{I}_2$$
$$\mathcal{E}_2 = -\dot{\Phi}_2 = N_2 \dot{\Phi} = -L_2 \dot{I}_2 - M \dot{I}_1$$

A5 — 0.50 pts

Obtain an expression for the current transformation ratio $k_I \equiv \left|\tilde{I}_1\right|/\left|\tilde{I}_2\right|$. The answer may include L_1, L_2, ω and Z.

Using the equations from A4 and the following relations:

$$U_2 = I_2 Z M^2 = L_1 L_2$$

$$I_2 Z = \mathcal{E}_2 = -L_2 \dot{I}_2 - M \dot{I}_1 I_2 \left(Z - i\omega L_2\right) = i\omega M I_1$$

we obtain

Answer: Answer:

$$k_I = \left|\frac{\tilde{I}_1}{\tilde{I}_2}\right| = \left|\frac{Z - i\omega L_2}{i\omega M}\right| = \frac{\sqrt{(\mathrm{Re}\, Z)^2 + (\mathrm{Im}\ Z - \omega L_2)^2}}{\omega\sqrt{L_1 L_2}}$$

A6 — 0.50 pts

What is the voltage transformation ratio $k_U = \left|\tilde{U}_2\right|/\left|\tilde{U}_1\right|$? The answer may include L_1, L_2, ω and Z.

Writing down the equations from the previous paragraphs:

$$U_2 = -L_2 \dot{I}_2 - M \dot{I}_1 U_2 = I_2 \cdot Z U_1 = L_1 \dot{I}_1 + M \dot{I}_2$$

express U_2 through U_1:

$$U_2 = i\omega L_2 \frac{U_2}{Z} + i\omega M I_1 U_1 = -i\omega L_1 I_1 - i\omega M \frac{U_2}{Z} U_2 \left(1 - \frac{i\omega L_2}{Z}\right) = \frac{M}{L_1}\left(U_1 - \frac{i\omega M}{Z} U_2\right) \frac{U_2}{U_1} = \frac{\frac{M}{L_1}}{1 - \frac{i\omega L_2}{Z} + \frac{i\omega M^2}{Z L_1}} = \frac{M}{L_1} = -$$

we get:

Answer: Answer:

$$k_U = \left|-\frac{N_2}{N_1}\right| = \sqrt{\frac{L_2}{L_1}}$$

A7 — 0.30 pts

Write down the condition for the number of turns in windings N_1 and N_2, under which the expression (1) is satisfied.

Answer: Answer:

$$N_2 \gg N_1$$

A8 — 0.60 pts

Determine the phase difference $\Delta\varphi$ by which voltage U_1 leads current I_1. The answer may include L_1, L_2, ω and Z.

$$U_1 + \frac{i\omega M}{Z} \cdot U_2 = -i\omega L_1 I_1$$

$$U_1 \left(1 - \frac{i\omega M N_2}{Z N_1}\right) = -i\omega L_1 I_1$$

$$U_1 = \frac{-i\omega L_1 Z N_1}{Z N_1 - i\omega M N_2} I_1$$

after transformations:

Or, which is convenient for further calculations,

$$\cos\Delta\varphi = \frac{\omega L_2 \mathrm{Re}\ Z}{|Z|\sqrt{(\mathrm{Re}\ Z)^2 + (\omega L_2 - \mathrm{Im}\ Z)^2}}$$

Answer: Answer:

$$\mathrm{tg}\Delta\varphi = \frac{|Z|^2 - \omega L_2 \cdot \mathrm{Im}\ Z}{\omega L_2 \mathrm{Re}\ Z}$$

A9 — 0.60 pts

Determine the time-averaged power P of source U_1. The answer may include U_0, Z, N_1 and N_2.

The average power is found as:

$$P = \frac{1}{2}|I_1||U_1|\cos\Delta\varphi$$

Substituting we get:

The calculations could be significantly reduced by using the following formula for power:

$$P = \mathrm{Re}\left(\frac{1}{2}I_1 U_1^*\right)$$

Answer: Answer:

$$P = \frac{U_1^2 \mathrm{Re}\ Z}{2|Z|^2} \cdot \frac{N_2^2}{N_1^2}$$

B1 — 0.50 pts

Find the ratio I/I_A.

Using the formula for k_I we get the answer:

Answer: Answer:

$$\frac{I}{I_A} = \sqrt{\frac{L_1}{L_2}} = N$$

C1 — 0.20 pts

Obtain an expression for the average power P_0, which is released at the load if the winding resistance can be neglected. Express your answer through the amplitude of the input voltage U_0 and the load resistance R. Calculate it for a transformer whose parameters were given in the problem earlier.

Using the formula for average power:

Answer: Answer:

$$P_0 = \frac{U_0^2}{2R} = \frac{U_{\mathrm{rms}}^2}{R} = 303\ \mathrm{W}$$

C2 — 0.30 pts

Similar to Part A, write equations relating the complex amplitudes $\tilde{U}_1$, $\tilde{U}_2$, $\tilde{I}_1$ and $\tilde{I}_2$. The response may also include R, r, single winding inductance L, and cyclic frequency ω.

Answer: Answer:

$$\tilde{U}_1 = -i\omega L\tilde{I}_1 + i\omega L\tilde{I}_2 + r\tilde{I}_1\tilde{U}_2 = (R+r)\tilde{I}_2\tilde{U}_2 = i\omega L\tilde{I}_2 - i\omega L\tilde{I}_1$$

C3 0.40 pts

Express $\tilde{I}_1$ and $\tilde{I}_2$ in terms of $\tilde{U}_1$ and other quantities you need, which may include R, r, L and ω.

We express:

$$\tilde{I}_2(R + r - i\omega L) = -i\omega L\tilde{I}_1$$

$$\tilde{U}_1 = \left(-i\omega L + r + \frac{\omega^2 L^2}{R + r - i\omega L}\right)\tilde{I}_1$$

$$\tilde{I}_1 = \frac{(R + r - i\omega L)\tilde{U}_1}{(r - i\omega L)(R + r - i\omega L) + \omega^2 L^2}$$

we get:

Answer: Answer:

$$\tilde{I}_1 = \frac{R + r - i\omega L}{r(R + r) - i\omega L(R + 2r)}\tilde{U}_1$$

$$\tilde{I}_2 = \frac{-i\omega L}{r(R + r) - i\omega L(R + 2r)}\tilde{U}_1$$

C4 1.10 pts

Hereinafter, work in the limit $r \ll R, \omega L$. Obtain expressions for the power of Joule losses P_{r1} and P_{r2} released in the primary and secondary windings. Express your answer in terms of the amplitude of the input voltage U_0, as well as r, ω and L. Calculate the total power of Joule losses on windings $P_r = P_{r1} + P_{r2}$ for a transformer, the parameters of which were given earlier in the problem. Provide detailed calculations for your calculations in the solution sheets.

Obtain expressions for the power of Joule losses P_{r1} and P_{r2} released in the primary and secondary windings. Express your answer in terms of the amplitude of the input voltage U_0, as well as r, ω and L.

Calculate the total power of Joule losses on windings $P_r = P_{r1} + P_{r2}$ for a transformer whose parameters are given earlier in the problem. Provide detailed calculations for your calculations in the solution sheets.

Let's write down the power released on the first and second windings:

$$P_{r1} = \frac{|\tilde{I}_1|^2 r}{2} = \frac{U_0^2 r}{2} \cdot \left|\frac{R + r - i\omega L}{r(R + r) - i\omega L(R + 2r)}\right|^2$$

$$P_{r2} = \frac{|\tilde{I}_2|^2 r}{2} = \frac{U_0^2 r}{2} \cdot \left|\frac{-i\omega L}{r(R + r) - i\omega L(R + 2r)}\right|^2$$

Note, that $r \ll R, \omega L$ we get:

Then the total power losses:

$$P_r = \frac{U_0^2 r}{2} \cdot \frac{R^2 + 2\omega^2 L^2}{\omega^2 L^2 R^2}$$

Let's do the calculations:

$$U_0 = U_{\text{rms}} \cdot \sqrt{2} = 311 \text{ V}, \quad \omega = 2\pi f = 314 \ \frac{\text{rad}}{s}, \quad L = \frac{\mu\mu_0 S N^2}{l} = 0.503 \text{ H}$$

Length of one turn:

$$c = 2\pi \cdot \sqrt{\frac{S}{\pi}} = 35.4 \text{ cm}$$

Resistance of winding:

$$r = \frac{1}{\sigma_{\text{w}}} \cdot \frac{Nc}{\frac{\pi d^2}{4}} = 2.16 \ \Omega$$

then

Answer: Answer:

$$P_{r1} = \frac{U_0^2 r}{2} \cdot \frac{R^2 + \omega^2 L^2}{\omega^2 L^2 R^2}, \qquad P_{r2} = \frac{U_0^2 r}{2R^2}$$

D1 — 0.30 pts

Write down the expression for the tangential electric field $E(r,t)$.

Let us write down the magnetic field flux through a circle of radius r:

$$\Phi = B\pi r^2$$

Law of electromagnetic induction for a circle of radius r:

$$\mathcal{E} = -\dot{\Phi} 2\pi r E(r,t) = i\omega\pi r^2 B_0 e^{-i\omega t}$$

Answer: Answer:

$$E(r,t) = \text{Re}\left(\frac{B_0\omega r}{2} e^{\frac{i\pi}{2} - i\omega t}\right) = \frac{B_0 \omega r}{2} \sin\omega t$$

D2 — 0.70 pts

Derive an expression for the average power of Joule losses P_{eddy} released in the core. Express your answer in terms of the conductivity of the core material σ_{c}, its cross-sectional area S, length ℓ, and B_0 and f.

Let us write down the power of Joule losses per unit volume:

$$p_{\text{eddy}}(r,t) = \sigma_c E^2 = \frac{\sigma_c B_0^2 \omega^2 r^2}{4} \sin^2 \omega t$$

average by time obtain:

$$\overline{p_{\text{eddy}}} = \frac{\sigma_c B_0^2 \omega^2 r^2}{8}$$

Then the mean power released in the entire core:

$$P_{\text{eddy}} = \frac{\sigma_c B_0^2 \omega^2 l}{8} \cdot \int_0^{\sqrt{\frac{S}{\pi}}} 2\pi r^3 dr$$

calculating

Answer: Answer:

$$P_{\text{eddy}} = \frac{\pi}{4} \cdot \sigma_c B_0^2 f^2 S^2 l$$

D3 — 0.30 pts

Find the factor A.

Let us write the law of electromagnetic induction for a rectangle with dimensions $2z \times 2x$:

$$E_x(z,t) \cdot 4x = -\dot{\Phi} = 2x \cdot 2z \cdot \text{Re}\ (i\omega B_0 e^{-i\omega t}) E_x(z,t) = \text{Re}\ (i\omega B_0 z e^{-i\omega t})$$

Answer: Answer:

$$A = i\omega B_0$$

D4 **1.40 pts**

Write down an expression for the power $\mathrm{d}P/\mathrm{d}V$ released in such a plate per unit volume, averaged over the entire height of the plate, as well as over time. Responses may include S, ℓ, B_0, h, σ_c and f. Let now the core be assembled from plates with the parameters indicated above in the problem. The load R is still connected to the secondary winding. Express the power P'_eddy that would be released in the core due to Foucault currents, in terms of the parameters given in the problem. Calculate this power.

Let's write down the expression for instantaneous power at a certain point per unit area:

$$\frac{\mathrm{d}P}{\mathrm{d}V}(z,t) = \sigma_c E^2(z,t) = \sigma_c \omega^2 B_0^2 z^2 \sin^2 \omega t$$

average by time obtain:

$$\frac{\mathrm{d}P}{\mathrm{d}V}(z) = \frac{1}{2}\sigma_c \omega^2 B_0^2 z^2$$

now average by height z:

$$\frac{\mathrm{d}P}{\mathrm{d}V} = \frac{1}{2}\sigma_c \omega^2 B_0^2 \cdot \frac{1}{h}\int_{-h/2}^{h/2} z^2 dz = \frac{1}{24}\sigma_c \omega^2 B_0^2 h^2$$

Let's find an expression for the power allocated to the entire transformer:

$$P'_{eddy} = \frac{\mathrm{d}P}{\mathrm{d}V} \cdot Sl = \frac{\pi^2}{6}\sigma_c B_0^2 f^2 h^2 Sl$$

one can note that here the power is proportional to S, and not S^2 (as would be the case for a solid core), which means power losses will be significantly lower. Now find the amplitude of the magnetic field B_0. Use the equation relating the magnetic flux through the core and the voltage on the winding from part A:

$$U_1 = N\dot{\Phi} = NS(-i\omega B)B_0 = \frac{U_0}{2\pi f NS} = 0.99 \text{ T}$$

calculate P'_eddy:

Now let's find the amplitude of the magnetic field B_0. We use the equation to relate the magnetic field flux through the core and the voltage on the winding from part A:

$$U_1 = N\dot{\Phi} = NS(-i\omega B)B_0 = \frac{U_0}{2\pi f NS} = 0.99 \text{ T}$$

calculate P'_eddy:

Answer: Answer:

$$\frac{\mathrm{d}P}{\mathrm{d}V} = \frac{\pi^2}{6}\sigma_c B_0^2 f^2 h^2$$

E1 **0.40 pts**

For simplicity, consider a transformer without load. Record the energy that the source imparts to the core when current I flows through the primary winding and the flux through the core changes to $\mathrm{d}\Phi$. Then express this energy in terms of H, $\mathrm{d}B$ and core parameters.

Answer: Answer:

$$\mathrm{d}W = N \cdot I\,\mathrm{d}\Phi = N\frac{Hl}{N}S\,\mathrm{d}B = Sl \cdot H\,\mathrm{d}B$$

E2 **0.30 pts**

We will assume that the last expression is valid in an arbitrary case. Write down the expression for the average power P_{hyst} hysteresis losses. Express your answer in terms of H_{c}, B_{s} and the quantities given at the beginning of part C. Also get the numerical value.

For one period on the graph $B(H)$ one complete hysteresis loop will be described, which means

$$\Delta W = Sl \cdot \Delta H \Delta B = Sl \cdot 2H_c \cdot 2B_s = 4SlH_cB_s$$

Then the mean power lost due to hysteresis is equal to:

$$P_{\text{hyst}} = \frac{\Delta W}{T}$$

energy will be lost

Answer: Answer:

$$P_{\text{hyst}} = 4SlH_cB_sf = 15 \text{ W}$$

E3 **0.50 pts**

Assuming that all contributions calculated in parts C, D and E are small and independent, calculate the final efficiency of the transformer η.

$$\eta = 1 - \frac{P_r + P_{\text{eddy}} + P_{\text{hyst}}}{P_0}$$

Answer: Answer:

$$\eta = 91 \text{ \%}$$